a

Task

Colour fundus photograph in, four lesion probability maps out



- **MA** microaneurysm
- **HE** hemorrhage
- **EX** hard exudate
- **SE** soft exudate

Data collection

8 test sets, 7 datasets · 5,121 images

Training sources	
IDRiD	81
DDR	757
Independent sources	
TJDR-std	257
TJDR-uwf	304
MAPLES-DR	198
FGADR	1,842
Retinal-Lesions	1,593
DiaRetDB1	89

Model collection

39 variants in the registry

- Tier 1 · main table — 22
- Tier 2 · later rounds — 6
- Tier 3 · paper audit only — 11

Reproducibility level R0–R4 recorded for every variant; failures stay in the denominator

Four evaluations

- **Paper audit** — numbers as reported, and where they came from
- **Released checkpoints** — scored with the paper's number, the authors' metric and ours
- **Author-recipe reruns** — retrained with the authors' own pipeline
- **Unified retraining** — shared data, budget, augmentation, selection

Tests and metrics

- **Internal test** — official test set of the training source
- **Independent test** — sources never used for training

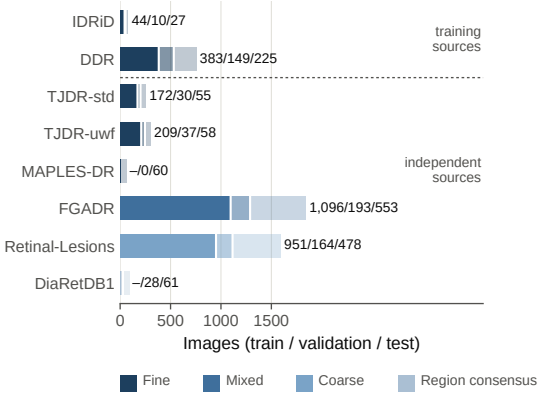
Primary — exact pooled AUPR inside the field of view; mAUPR

Secondary — Dice, IoU; lesion-level recall by lesion size

Statistics — 3 seeds; image bootstrap; Friedman + Nemenyi

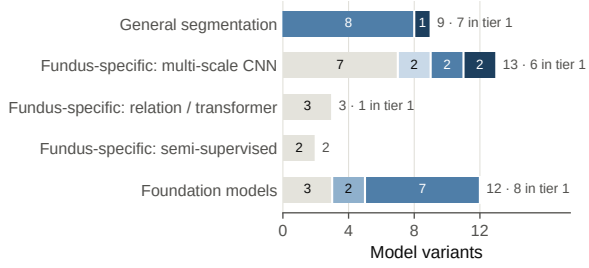
b

Test sources and frozen splits



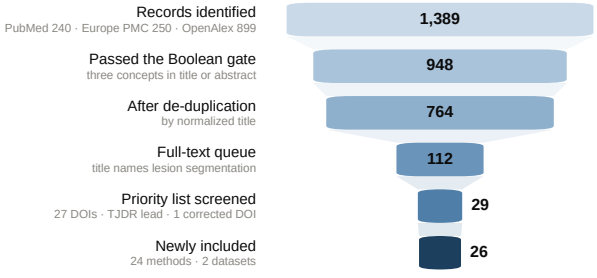
c

Registered models by family and reproducibility level



d

Literature search and screening



Fixed search 2015-01-01 to 2026-09-30; with the snowballed corpus the ledger holds 86 records.

e

Factors assessed

Training source — *5 representative models*: IDRiD · DDR · TJDR-std · FGADR	**Label granularity** — *U-Net · M2MRF*: fine · mixed · coarse, alone and combined	**Training amount** — *U-Net · M2MRF*: 10 · 50 · 100% of DDR; fine sources merged	**Input resolution** — *M2MRF · RETFound*: 512 · 1,440 canvas · native-resolution patches
Loss — *U-Net*: CE · Dice · CE + Dice · CE + per-image Dice	**Learning rate** — *HRNet-W48 · M2MRF*: recipe value × ⅓, 1, 3; validation only	**Foundation models** — *8 models*: zero-shot · oracle prompt PEFT · encoder + decoder	**Encoder pretraining** — *4 encoders*: retinal vs natural-image ViT-L, one UPerNet head

One factor changes at a time; everything else stays fixed. The five representative models are U-Net, nnU-Net, M2MRF, SAM2-UNet and RETFound + UPerNet.